Lab 9

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Task 1

Created URL `/books/lab9/task1` that lists all books with their percentage availability out of total stock, using a transient field on the `Book` model.

Files changed:

- `apps/bookmodule/models.py` — added `percentage` as a transient (non-DB) field on `Book`:

```
class Book(models.Model):
    ...
    percentage = None  # transient field, not stored in DB
```

- `apps/bookmodule/urls.py` — added route:

```
path('lab9/task1', views.lab9_task1, name="books.lab9_task1"),
```

- `apps/bookmodule/views.py` — added view at the bottom:

```
def lab9_task1(request):
    total = Book.objects.aggregate(total=Sum('quantity'))['total'] or
1
    mybooks = Book.objects.all()
    for book in mybooks:
        book.percentage = round((book.quantity / total) * 100, 2)
    return render(request, 'bookmodule/lab9_task1.html', {'books':
mybooks})
```

- `apps/templates/bookmodule/lab9_task1.html` — new template displaying each book's title, quantity, and availability percentage.

```
{% extends "layouts/base.html" %}
{% block title %} Lab9 Task1 {% endblock title %}
{% block content %}
<h1>Books Availability</h1>
{% for book in books %}
  Title: {{ book.title }}, Quantity: {{ book.quantity }},
Availability: {{ book.percentage }}%
  <hr>
{% empty %}
```

```
  <p>No books found.</p>
{% endfor %}
{% endblock content %}
```

Output at `http://localhost:8000/books/lab9/task1`:

Task 2

Created URL `/books/lab9/task2` that lists all publishers annotated with the sum of quantities of all their books using `annotate` and `Sum`.

- `apps/bookmodule/urls.py` — added route:

```
path('lab9/task2', views.lab9_task2, name="books.lab9_task2"),
```

- `apps/bookmodule/views.py` — imported `Publisher` and added view at the bottom:

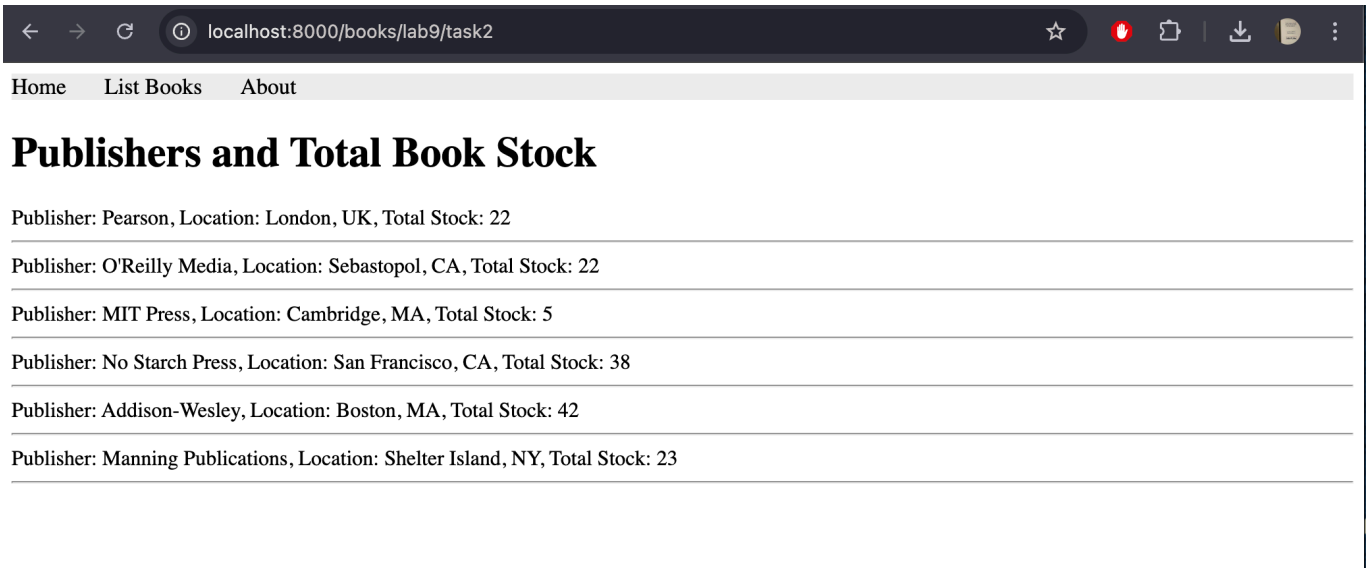
```
from .models import Book, Publisher

def lab9_task2(request):
```

```
    publishers =
Publisher.objects.annotate(total_stock=Sum('book__quantity'))
    return render(request, 'bookmodule/lab9_task2.html',
{'publishers': publishers})
```

- `apps/templates/bookmodule/lab9_task2.html` — new template displaying each publisher's name, location, and total stock.

```
{% extends "layouts/base.html" %}
{% block title %} Lab9 Task2 {% endblock title %}
{% block content %}
<h1>Publishers and Total Book Stock</h1>
{% for publisher in publishers %}
  Publisher: {{ publisher.name }}, Location: {{ publisher.location }},
Total Stock: {{ publisher.total_stock }}
  <hr>
{% empty %}
  <p>No publishers found.</p>
{% endfor %}
{% endblock content %}
```

Output at `http://localhost:8000/books/lab9/task2`:



Task 3

Created URL `/books/lab9/task3` — the opposite of Task 2. Instead of total stock, it retrieves the oldest book (earliest `pubdate`) managed by each publisher using `Subquery` with `OuterRef`.

- `apps/bookmodule/urls.py` — added route:

```
path('lab9/task3', views.lab9_task3, name="books.lab9_task3"),
```

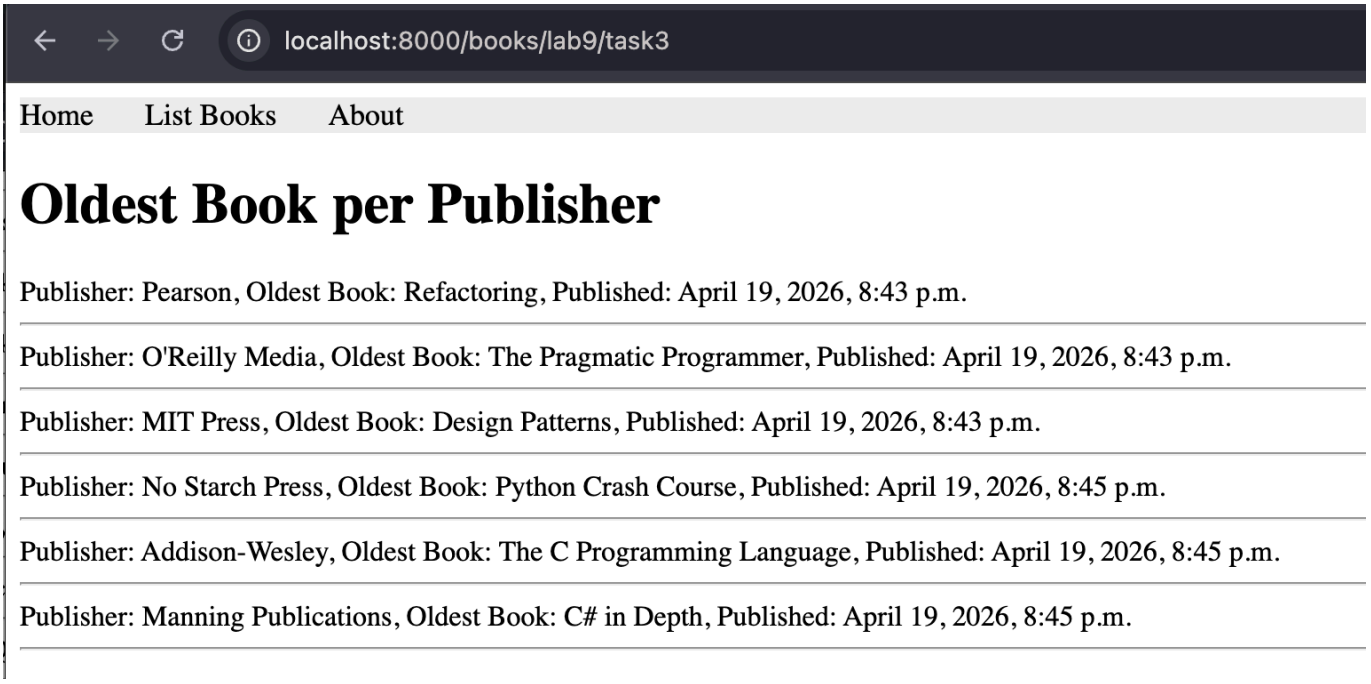
- `apps/bookmodule/views.py` — imported `Subquery` and `OuterRef` and added view at the bottom:

```
from django.db.models import Q, Count, Sum, Avg, Max, Min, Subquery,
OuterRef

def lab9_task3(request):
    oldest_title = Book.objects.filter(
        publisher=OuterRef('pk')
    ).order_by('pubdate').values('title')[:1]

    oldest_pubdate = Book.objects.filter(
        publisher=OuterRef('pk')
    ).order_by('pubdate').values('pubdate')[:1]

    publishers = Publisher.objects.annotate(
        oldest_book=Subquery(oldest_title),
        oldest_pubdate=Subquery(oldest_pubdate),
    )
    return render(request, 'bookmodule/lab9_task3.html',
{'publishers': publishers})
```

- `apps/templates/bookmodule/lab9_task3.html` — new template displaying each publisher's name, their oldest book title, and its publication date.

```
{% extends "layouts/base.html" %}
{% block title %} Lab9 Task3 {% endblock title %}
{% block content %}
<h1>Oldest Book per Publisher</h1>
{% for publisher in publishers %}
  Publisher: {{ publisher.name }}, Oldest Book: {{
publisher.oldest_book }}, Published: {{ publisher.oldest_pubdate }}
  <hr>
{% empty %}
  <p>No publishers found.</p>
{% endfor %}
{% endblock content %}
```

Output at `http://localhost:8000/books/lab9/task3`:



Task 4

Created URL `/books/lab9/task4` that calculates average, min, and max price of books for each publisher using `annotate` with `Avg`, `Min`, and `Max`.

- `apps/bookmodule/urls.py` — added route:

```
path('lab9/task4', views.lab9_task4, name="books.lab9_task4"),
```

- `apps/bookmodule/views.py` — added view at the bottom:

```
def lab9_task4(request):
    publishers = Publisher.objects.annotate(
        avg_price=Avg('book__price'),
        min_price=Min('book__price'),
        max_price=Max('book__price'),
    )
    return render(request, 'bookmodule/lab9_task4.html',
{'publishers': publishers})
```

- `apps/templates/bookmodule/lab9_task4.html` — new template displaying each publisher's price stats.

```
{% extends "layouts/base.html" %}
{% block title %} Lab9 Task4 {% endblock title %}
{% block content %}
<h1>Book Price Stats per Publisher</h1>
{% for publisher in publishers %}
```

```
  Publisher: {{ publisher.name }}, Avg Price: {{
publisher.avg_price|floatformat:2 }}, Min Price: {{
publisher.min_price }}, Max Price: {{ publisher.max_price }}
  <hr>
{% empty %}
  <p>No publishers found.</p>
{% endfor %}
{% endblock content %}
```

Output at `http://localhost:8000/books/lab9/task4`:

Task 5

Created URL `/books/lab9/task5` that lists publishers with a count of their highly rated books (rating ≥ 4), using `Count` with a `filter` argument.

- `apps/bookmodule/urls.py` — added route:

```
path('lab9/task5', views.lab9_task5, name="books.lab9_task5"),
```

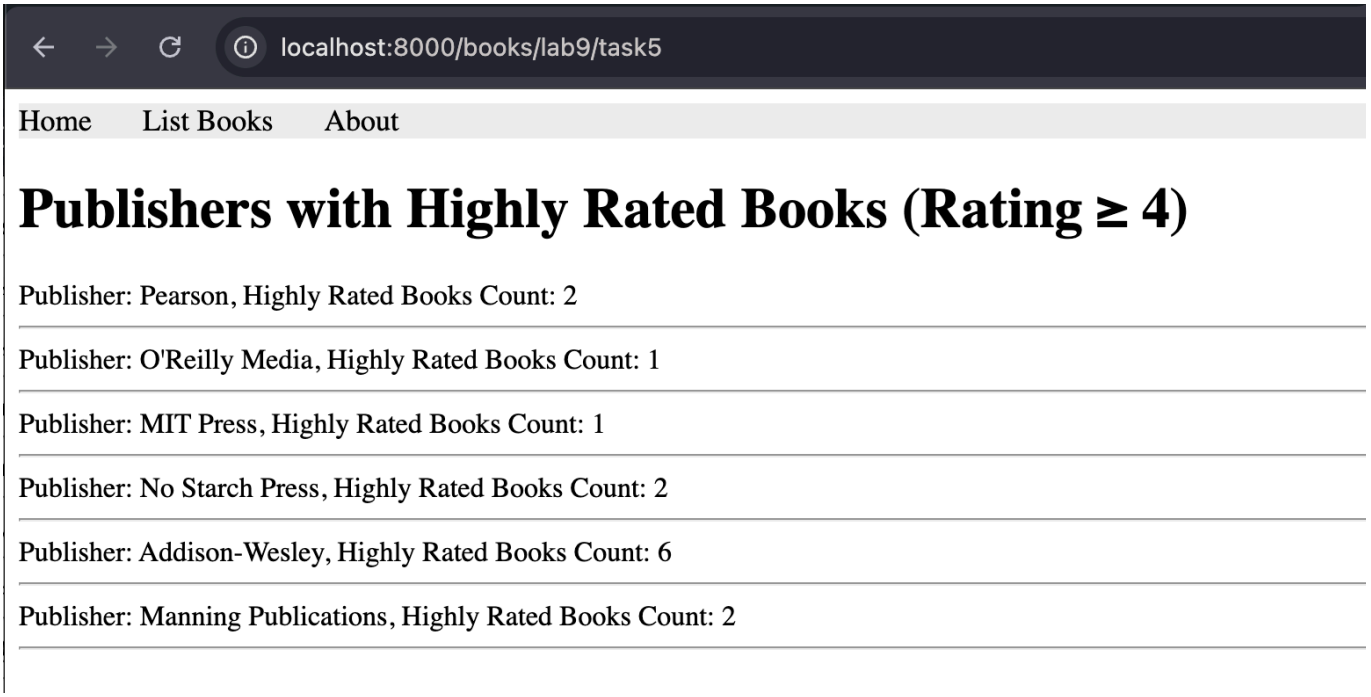
- `apps/bookmodule/views.py` — added view at the bottom:

```
def lab9_task5(request):
    publishers = Publisher.objects.annotate(
        highly_rated_count=Count('book',
filter=Q(book__rating__gte=4))
    ).filter(highly_rated_count__gt=0)
    return render(request, 'bookmodule/lab9_task5.html',
{'publishers': publishers})
```

- `apps/templates/bookmodule/lab9_task5.html` — new template displaying each publisher's name and highly rated book count.

```
{% extends "layouts/base.html" %}
{% block title %} Lab9 Task5 {% endblock title %}
{% block content %}
<h1>Publishers with Highly Rated Books (Rating &ge; 4)</h1>
{% for publisher in publishers %}
  Publisher: {{ publisher.name }}, Highly Rated Books Count: {{
publisher.highly_rated_count }}
  <hr>
{% empty %}
  <p>No publishers found.</p>
{% endfor %}
{% endblock content %}
```

Output at `http://localhost:8000/books/lab9/task5`:



Task 6

Created URL `/books/lab9/task6` that lists publishers with a count of their books priced > 50 and quantity between 1 and 4 (inclusive), using `Count` with a combined `Q` filter.

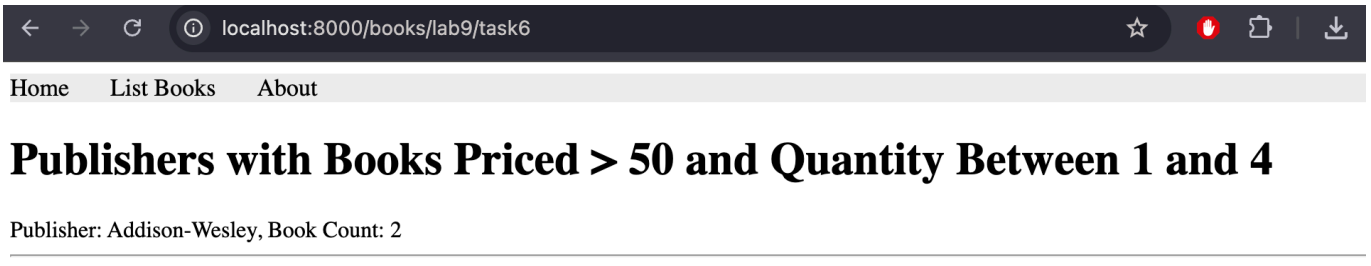
- `apps/bookmodule/urls.py` — added route:

```
path('lab9/task6', views.lab9_task6, name="books.lab9_task6"),
```

- `apps/bookmodule/views.py` — added view at the bottom:

```
def lab9_task6(request):
    publishers = Publisher.objects.annotate(
        book_count=Count('book', filter=Q(book__price__gt=50) &
Q(book__quantity__gte=1) & Q(book__quantity__lt=5))
    ).filter(book_count__gt=0)
    return render(request, 'bookmodule/lab9_task6.html',
{'publishers': publishers})
```

- `apps/templates/bookmodule/lab9_task6.html` — new template displaying each publisher's name and filtered book count.

```
{% extends "layouts/base.html" %}
{% block title %} Lab9 Task6 {% endblock title %}
{% block content %}
<h1>Publishers with Books Priced &gt; 50 and Quantity Between 1 and
4</h1>
{% for publisher in publishers %}
  Publisher: {{ publisher.name }}, Book Count: {{ publisher.book_count
}}
  <hr>
{% empty %}
  <p>No publishers found.</p>
{% endfor %}
{% endblock content %}
```

Output at `http://localhost:8000/books/lab9/task6`: